1.Free Use vs Paid API

Direct links / embedded widgets

- Think of this like giving students a **button** in your SmartCampus portal that opens ChatGPT or Perplexity in a new window.
- **Cost:** $0 for you, $0 for students.
- **Limitations:** Students can only use what's available in the free tier (GPT-3.5, basic search). No automation inside SmartCampus.

API integration

- This is when you connect ChatGPT or Perplexity **directly to your SmartCampus backend**. Students/teachers interact without leaving your portal.
- **Cost:** You pay per usage. OpenAI charges per token (a "token" is roughly 4 characters of text), and Perplexity Pro requires a subscription.
- **Benefit:** Full automation: summaries, grading suggestions, analytics, dashboards, internal workflows.

2.Cost Illustration

Let's do a **step-by-step calculation** for OpenAI GPT-3.5:

1. **Assumptions:**
   - 500 students
   - 10 queries per student per day
   - Average query = 200 tokens (input + output)
   - School month = 20 school days
2. **Total tokens per month:**
   - 500 students × 10 queries/day × 200 tokens/query × 20 days
   - Step by step:
     - 500 × 10 = 5,000 queries/day
     - 5,000 × 200 = 1,000,000 tokens/day
     - 1,000,000 × 20 days = **20,000,000 tokens/month**
3. **Cost (GPT-3.5 @ $0.002 per 1,000 tokens):**
   - 20,000,000 tokens ÷ 1,000 = 20,000 units
   - 20,000 × $0.002 = **$40/month**

 So, for 500 students using GPT-3.5 moderately, the monthly API cost is about $40.

- GPT-4 would be 15–30× more expensive ($600–$1,200/month) for the same usage.

3.Key Takeaways

Factor	Free Links / Embeds	API Integration
Cost for School	$0	Paid (scales with usage)
Cost for Students	$0	$0 (absorbed by school)
Automation	None	Summaries, grading, analytics
Feature Limitations	Limited (GPT-3.5 / basic)	Full features (GPT-4, analytics)
Compliance / Privacy	Minimal (students go outside portal)	Must handle student data responsibly

---

4.Smart Strategy (Hybrid Approach)

- **Students:** Free links/embeds → zero cost, full access to GPT-3.5.
- **Teachers/Admins:** API integration → enable automation and productivity features where the cost is justified.